FINAL CREDIT RISK REPORT:

A One-Year Comparative Analysis of Credit Risk in Two Portfolios

Submitted to the
Chair of Finance

Introduction to Credit Risk and Applications in Python

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1 Introduction

This report presents a one-year credit risk analysis of two loan portfolios, Portfolio 1 and Portfolio 2. Five datasets were provided in CSV format. Following a thorough data cleaning process, this report first provides descriptive statistics of both portfolios, establishing initial intuitions about their relative riskiness. Two simulation-based default models are then applied: the Gaussian one-factor model and the beta-binomial model. The main objective is to compare three common risk measures: Expected loss (EL), Value-at-Risk (VaR), and Expected shortfall (ES), across both portfolios and both models, justifying parametric choices and explaining differences in results that arise from underlying assumptions.

2 Data Cleaning

The first and most critical step was preparing a complete dataset suitable for modeling. Since both the Gaussian one-factor and beta-binomial simulations require valid estimates of probability of default (PD), loss given default (LGD), and exposure at default (EAD) for every obligor, particular attention was given to these variables and to entity matching across source files.

Entity identifiers were stripped of ”empty spaces” and converted to a uniform uppercase format to ensure correct matching across all datasets. In Portfolio 2, four entity codes were found to be missing the expected hyphen separator (e.g. `BSQ4OLC` instead of `B-SQ4OLC`). These were corrected manually. Also, all date fields were converted to a consistent datetime format. Duplicate observations for the same entity on the same date were removed. Missing values in the `default_event` field were replaced with zero, reflecting the assumption that the absence of a recorded default event implies no observed default at that point in time.

The industry labels exhibited several inconsistencies (e.g. `industry a`, `Ind. A`, `IndustryA`). These were mapped to labels (`Industry A`, `Industry B`). The `irrelevant_score` column was dropped as it carries no relevance to the subsequent analyses. Duplicate entity records were removed, retaining only the first occurrence of each entity code and industry returns were chronologically ordered.

All datasets were merged using left joins to preserve all portfolio observations, including those with missing data in the reference column. Entity information was merged first, followed by default status. A “latest default” table was constructed by selecting the most recent default history record per entity. During the modeling phase, the full default history became relevant for the calibration in the beta distribution, where annual default frequencies were aggregated to estimate industry-specific default rates, but more on that later.

Finally, the expected loss formula,

$$EL = PD \times LGD \times EAD, \tag{1}$$

requires valid values for all three components. Observations missing any of EAD, LGD, or PD were therefore excluded. Both portfolios originally contained 30 unique observations after deduplication. After removing rows with missing EAD and subsequently rows with missing LGD or PD, each portfolio retained 26 obligors for all subsequent analyses.

```
Raw rows:                  30
After removing NaN EAD:    28
After removing NaN LGD/PD: 26
```

3 Descriptive Statistics

Portfolio 1 exhibits a lower average PD of 1.99% compared with 3.46% for Portfolio 2, indicating stronger overall credit quality. The average LGD is also lower in Portfolio 1 (46.2% vs. 50.2%), implying better expected recovery in the event of default. In terms of total exposure, Portfolio 1 has an aggregate EAD of approximately $598 million, while Portfolio 2 carries a larger exposure of $678 million.

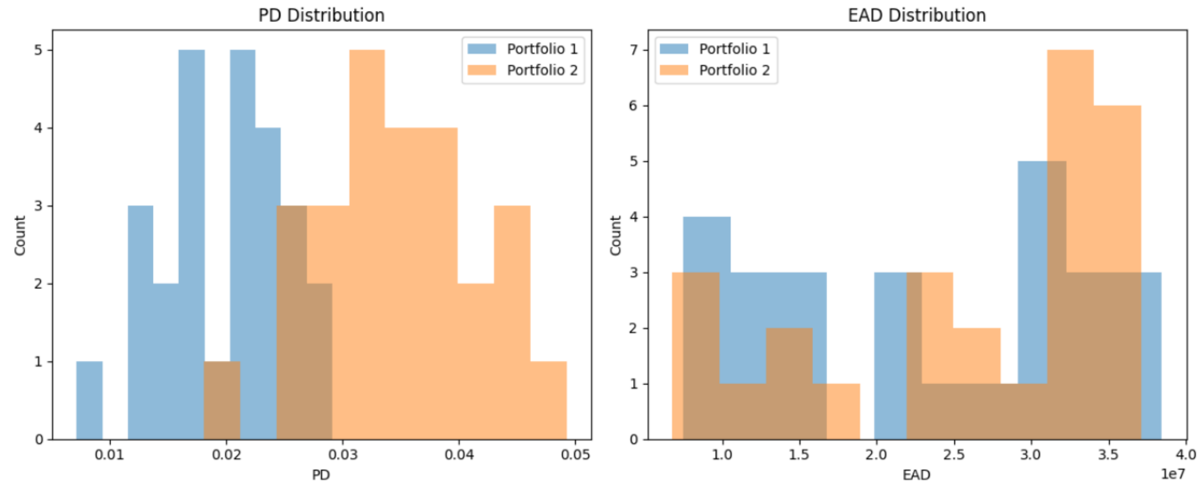

Figure 1: Distribution of PD and EAD

The PD histogram (Figure 1) shows a clear rightward shift for Portfolio 2, with its entire distribution lying above that of Portfolio 1. Notably, the minimum PD in Portfolio 2 (1.81%) exceeds the mean PD of Portfolio 1 (1.98%), meaning even the least risky obligor in Portfolio 2 is comparable to the average obligor in Portfolio 1. The EAD distributions are broadly similar across portfolios, both exhibiting a wide spread of individual exposures and no dominant single obligor.

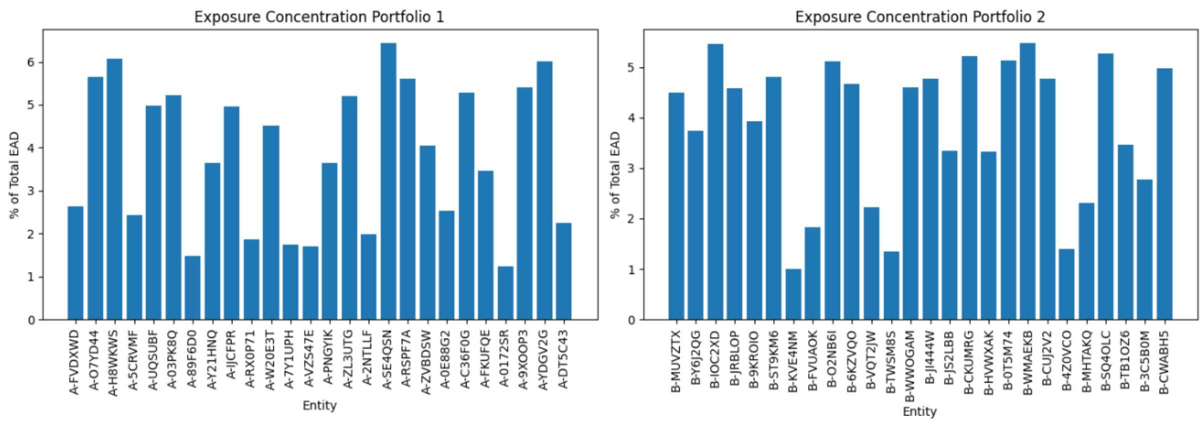

Figure 2: Exposure concentration by obligor for Portfolio 1 and Portfolio 2.

The exposure concentration (Figure 2) confirms that both portfolios are reasonably well diversified, with no single entity accounting for a disproportionate share of total EAD. Differences in portfolio-level risk measures are therefore more likely to reflect credit quality and dependence structure than exposure concentration.

The key structural distinction observed in the data between the two products is their industry composition: Portfolio 1 consists exclusively of entities from Industry A, while Portfolio 2 consists exclusively of entities from Industry B. The descriptive evidence therefor suggests that Industry B is inherently riskier. However, default dependence, systemic risk, or loss clustering are not captured by these simple observations. The following sections use the Gaussian one-factor and Beta-Binomial models to investigate whether these differences translate into higher expected losses and tail-risk measures under alternative dependence assumptions.

4 Model Theory

Before implementing the simulations, it is crucial to recall the theoretical foundations they build on. The two models differ fundamentally in how they form dependence. The Gaussian one-factor model generates it through a shared systematic factor F controlled by the deterministic parameter ρ. The beta-binomial model generates dependence through stochastic variation in the default rate, empirically calibrated from historical default frequencies. For both models, three risk measures are computed: Expected Loss (EL), Value-at-Risk at 99% confidence (VaR_{99}), and Expected Shortfall at 99% (ES_{99}). EL measures the average portfolio loss over time, VaR captures the loss level not exceeded in 99% of scenarios, and ES is the average loss conditional on exceeding VaR (McNeil, Frey & Embrechts, 2015). EL captures central tendency and the two others tail behaviour of the loss distribution.

4.1 Gaussian one-factor model

As discussed throughout the seminar, in the Gaussian one-factor model each obligor's latent asset return is modelled as:

$$A_i = \sqrt{\rho} \cdot F + \sqrt{1-\rho} \cdot \varepsilon_i, \tag{2}$$

where $F \sim \mathcal{N}(0,1)$ is a common systematic factor shared across all obligors representing the state of the economy, and $\varepsilon_i \sim \mathcal{N}(0,1)$ is an idiosyncratic shock specific to obligor i. Default occurs when the asset return falls below a threshold derived from the obligor's PD:

$$D_i = \Phi^{-1}(PD_i). \tag{3}$$

The total portfolio loss in each simulation is:

$$L = \sum_{i=1}^{N} \mathbf{1}[A_i < D_i] \cdot EAD_i \cdot LGD_i. \tag{4}$$

Dependence across obligors is fully captured by ρ. Here a value of $\rho = 0.2$ is used, following the Basel II calibration (Fitch, 2008). A sensitivity analysis varying $\rho \in \{0.1, 0.2, 0.3\}$ is included to test robustness. It's results confirm that as ρ increases, VaR and ES rise substantially while EL remains approximately unchanged, consistent with the theoretical result that correlation primarily affects tail risk rather than average losses (McNeil, Frey & Embrechts, 2015) (see code for exact outputs).

The model was implemented via Monte Carlo simulation with 100,000 draws. The simulated EL is checked against the analytical formula $EL = \sum_i PD_i \cdot LGD_i \cdot EAD_i$, with agreement within 0.2%, pointing to correct modeling.

4.2 Beta-binomial model

The beta-binomial model provides a reduced-form alternative in which dependence arises through stochastic variation in the portfolio default probability rather than fixed correlation (McNeil, Frey & Embrechts, 2015). The portfolio-wide PD a random variable:

$$p \sim \text{Beta}(\alpha, \beta). \tag{5}$$

Depending on p, individual defaults or no default are independent Bernoulli trials:

$$D_i \mid p \sim \text{Bernoulli}(p). \tag{6}$$

The parameters α and β are estimated by fitting a beta distribution to annual historical default rates observed in each industry over the 20-year historic data given in one of the files. When p is drawn high, many obligors default simultaneously, capturing crisis clustering without an explicit asset correlation structure, as in the Gaussian model.

5 Results

5.1 Gaussian One-Factor Model

After running the Gaussian one-factor simulation, the resulting loss distributions are heavily right-skewed for both portfolios (Figure 3). The majority of simulated scenarios produce very low losses, reflecting the rarity of large default events for the whole economy. However, the right tail extends to very large monetary losses.

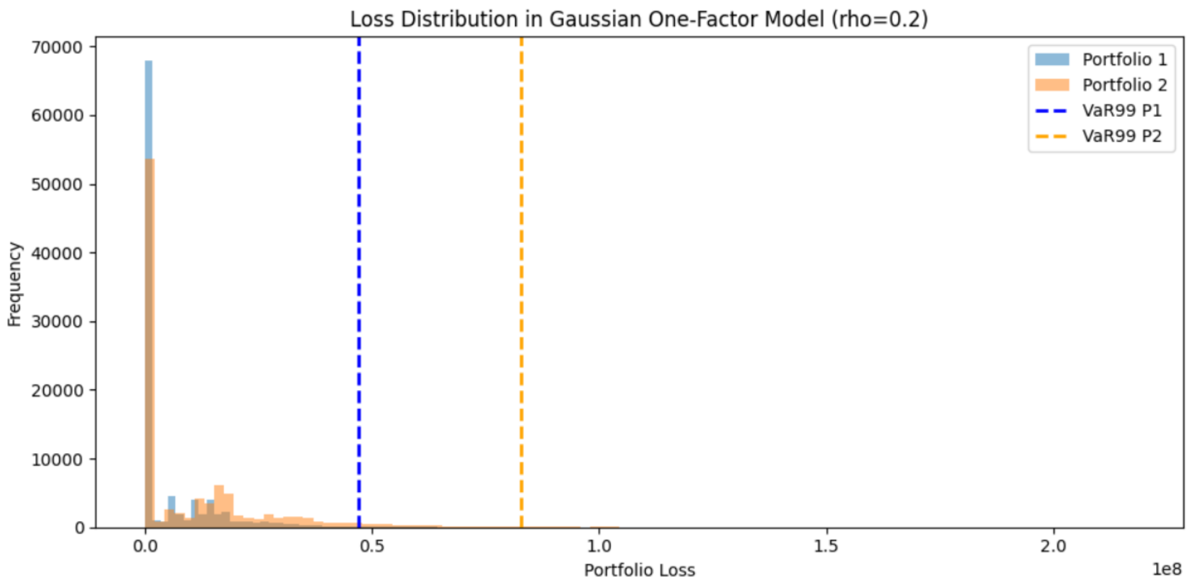

Figure 3: Simulated loss distributions under the Gaussian one-factor model ($\rho = 0.2$).

The VaR_{99} lines are clearly separated, with Portfolio 2 exhibiting a substantially higher threshold (\$83M vs.\$47M for Portfolio 1). This difference is primarily driven by Portfolio 2's higher average PD and LGD rather than by exposure differences, since EAD profiles are similar. The sensitivity analysis confirms that EL is stable across correlation assumptions, while VaR and ES increase sharply with ρ, and that Portfolio 2 is more sensitive to increasing correlation than Portfolio 1. This makes sense since higher individual PDs added with stronger co-movement amplify the probability of large simultaneous default events.

5.2 Beta-binomial model with the initial calibration

In the initial beta-binomial calibration, a single beta distribution was fitted to the pooled 20-year default history across all entities, yielding $\alpha = 1.3252$ and $\beta = 45.3824$ with a mean default rate of 2.84%. The small α relative to β reflects a right-skewed distribution concentrated near zero. However, further analysis suggested that the pooled calibration was inappropriate, as it ignores the structural differences between Industries A and B, which we hypothesised to be the primary source of variation between the two portfolios.

5.3 Model with industry specific calibration

Separating the historical data by industry reveals important differences in default behaviour. Industry A (Portfolio 1) exhibits a mean annual default rate of approximately 1.8%, with 10 out of 20 years recording zero defaults, thus long periods of calm punctuated by occasional spikes (forex: 2011: 8%). Industry B (Portfolio 2) has a higher mean default rate of approximately 3.9%, with more consistent annual default activity and a pike in 2020 (12.2%), likely related to the Covid pandemic stress. These differences strongly justify fitting separate beta distributions. The new parameters parameters are:

```
Portfolio 1 (Industry A): alpha = 0.3002, beta = 17.4797, mean = 1.69%
Portfolio 2 (Industry B): alpha = 0.8775, beta = 21.1289, mean = 3.99%
```

The very small α for Industry A reflects a highly dispersed Beta distribution, the default rate is either near zero or sometimes very high, with little in between.

5.4 Model comparison

The corrected results are presented in Table 1 and illustrated in Figure 5.

Table 1: Risk measures (values in millions).

Metric	P1 Gaus.	P2 Gaus.	P1 BB pooled	P2 BB pooled	P1 BB correct	P2 BB correct
EL	5.31	11.81	7.66	9.37	4.55	13.13
VaR 99%	46.9	82.9	48.0	58.2	51.4	81.5
ES 99%	61.3	102.9	58.5	71.1	66.6	98.9

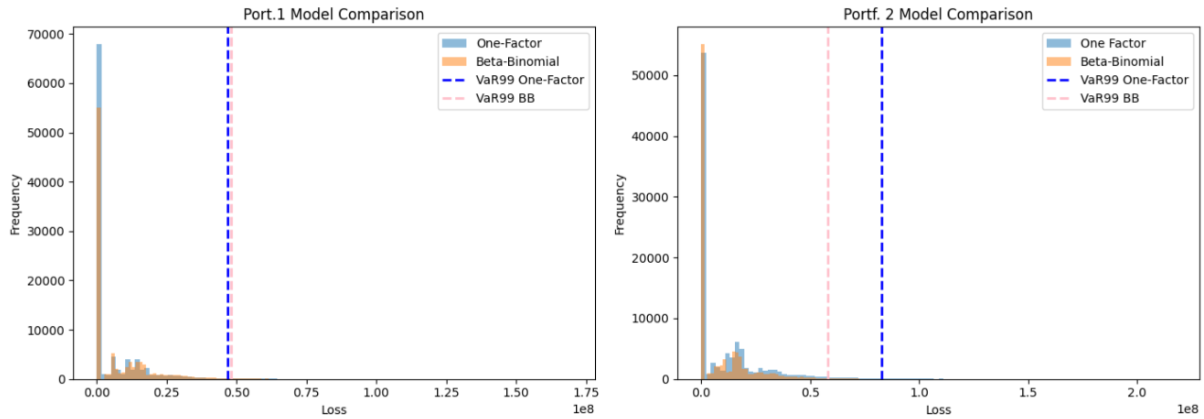

Figure 4: Loss distributions under the beta-binomial model (pooled calibration)

Two key findings arise. For Portfolio 2 model a convergence after correction is observed. The Gaussian and corrected beta-binomial models produce nearly identical VaR_{99} estimates:\$82.9M vs. \$81.5M. Industry B's fitted mean default rate of 3.99% aligns closely with Portfolio 2's base PD of 3.46%, and its historically more consistent default activity generates sufficient tail risk in the beta-binomial to match the $\rho = 0.2$ assumption in the Gaussian model. This cross-validates the tail risk estimates of Portfolio 2 across models.

For Portfolio 1, the corrected beta-binomial produces higher VaR_{99} and ES_{99} than the Gaussian model (\$51.4M vs. \$46.9M), despite a lower fitted mean default rate (1.69% vs. portfolio base PD of 1.99%). This divergence could be explained by the high variance in Industry A's historical default behaviour. The highly dispersed beta distribution ($\alpha = 0.30$) generates long periods of near-zero defaults followed by sharp spikes, which creates fatter tails in the beta-binomial

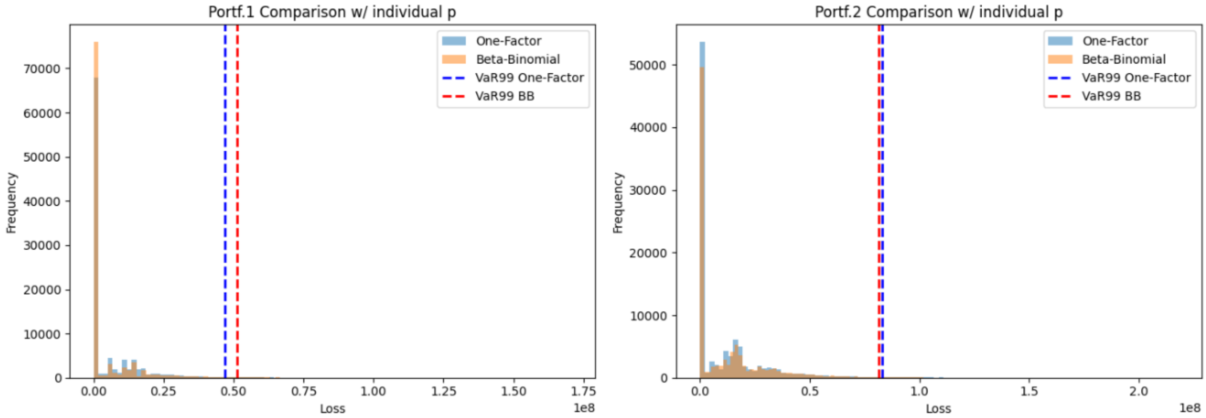

Figure 5: Loss distributions: One-factor model vs. corrected beta-binomial model.

loss distribution than the Gaussian model with fixed $\rho = 0.2$ can generate. The Gaussian model smooths this cluster behaviour through a constant correlation assumption, systematically underestimating tail risk for portfolios with historically volatile default dynamics.

6 Conclusion

In conclusion, across all metrics and both models, Portfolio 2 consistently exhibits higher expected losses and tail-risk measures than Portfolio 1, reflecting its higher average PD, higher LGD, and greater total exposure. Since Portfolio 1 and Portfolio 2 map exclusively to Industry A and Industry B respectively, this provides evidence that Industry B carries inherently higher credit risk.

The comparison of models reveals that neither dominates the other systematically. When parameters are calibrated to portfolio-specific historical data, the two models converge to consistent tail risk estimates for Portfolio 2, whose default behavior is relatively stable and persistent. For Portfolio 1 however, the beta-binomial produces significantly higher tail risk estimates, driven by Industry A's history of volatile and spiky default cycles.

The one-factor model with a fixed correlation parameter is well suited to portfolios with stable default dynamics, while the beta-binomial model benefits of greater flexibility in capturing higher variability and clustering in extreme losses. All in all, this shows that the choice of model and the calibration of its parameters to portfolio-specific historic data, has an important impact on risk measures and their subsequent interpretation.

References

Fitch Ratings: Hansen M., (2008). *Basel II correlation values: An empirical analysis of EL, UL and the IRB model*. Fitch Ratings

McNeil, A. J., Frey, R., & Embrechts, P. (2015). *Quantitative risk management: Concepts, techniques and tools* (Revised ed.). Princeton University Press